Feature Decoupling and Nonuniform Knowledge Transfer for Knowledge Distillation in Remote Sensing Image Object Detection

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***Abstract*—Deep learning-based object detection in remote sensing images often achieves high performance but has high model complexity, hindering edge deployment. Knowledge distillation (KD) offers a potential solution for model lightweighting. However, most existing distillation methods are designed for natural scene images and are not suitable for the spatial heterogeneity, semantic discrepancies, and scale diversity of satellite and aerial images (remote sensing imagery). Therefore, this article proposes a nonuniform KD based on feature decoupling (NKDFD). NKDFD incorporates spatial priors derived from 2-D Gaussian distributions during feature distillation to achieve spatial decoupling of foreground and background features. Independent feature adaptation modules are applied to the decoupled features, enabling nonuniform feature extraction and knowledge transfer for distinct attributes. This design mitigates potential detection pattern discrepancies between teacher and student models resulting from spatial heterogeneity. A Kolmogorov–Arnold network (KAN)-based semantic decoupling and selection module is used due to its strong nonlinear modeling capability, reducing interference from semantic discrepancies. A cross-scale feature interaction (CFI) module is developed based on multiscale feature decoupling to enhance multiscale feature learning. Extensive experiments on the DOTA, DIOR, RSOD, and VHR10 datasets demonstrate NKDFD's consistent improvements for different detectors (single-stage, two-stage, anchor-based, and anchor-free). Comparisons with 13 mainstream distillation methods on DOTA and RSOD show NKDFD's superiority. It achieves the highest mean average precision (mAP) of 37.7% (RetinaNet) and 44.8% (ATSS) on DOTA and 48.8% (RetinaNet) and 54.0% (ATSS) on RSOD. Further analysis validates the necessity of nonuniform processing, showing that distinct features (e.g., foreground/background) require independent adapters; otherwise, the output distribution is not statistically significantly different from that of random data.**

***Index Terms*—Feature decoupling, knowledge distillation (KD), nonuniform knowledge transfer, object detection, remote sensing images.**

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I. Introduction

DEEP learning-based object detection has made significant advancements in complex scene analysis due to its powerful feature representation ability. However, prevailing remote sensing detection models primarily depend on deep network architectures characterized by high parameter complexity, resulting in an imbalance between computational efficiency and detection performance. The high spatial, temporal, and spectral resolution of remote sensing data increases image dimensionality, causing a significant increase in inference latency per image. The limited computational resources available on airborne and spaceborne platforms present challenges for the real-time deployment of complex models [1]. Therefore, model compression techniques have been widely used to achieve a balanced optimization of accuracy, efficiency, and generalization ability. These encompass lightweight network design [2], [3], [4], [5], [6], [7], [8], [9], knowledge distillation (KD) [10], [11], [12], [13], [14], model pruning [15], [16], [17], [18], and parameter quantization [19], facilitating the real-time interpretation of remote sensing imagery.

KD compresses models by transferring knowledge from a high-performing teacher model to a compact student model while maintaining comparable accuracy. Feature distillation, a key KD approach, achieves this by constraining the student model to mimic the teacher's intermediate features, improving the compressed model's performance. Representative methods include channelwise KD (CWD) using the Kullback–Leibler (KL) divergence constraint [20], KD using Pearson's correlation coefficient (PKD) [21], and masked generative distillation (MGD), which masks portions of the teacher features to enhance the student features [22]. Although these methods have demonstrated notable improvements in natural scene object detection, using them for object detection in remote sensing images provides only marginal gains. This limitation is primarily attributed to the significant differences between

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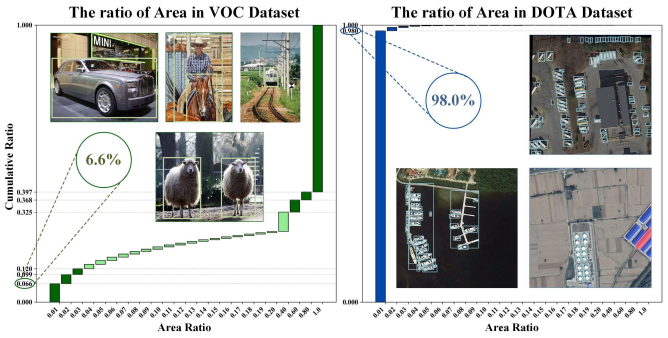


Fig. 1. Proportional area of objects at different scales in object detection datasets of remote sensing and natural scene images.

natural scene images and remote sensing imagery, as illustrated in Fig. 1. The primary reasons are as follows.

1) Remote sensing imagery exhibits significant spatial heterogeneity (as shown in Fig. 1, where dispersed and clustered objects coexist, whereas objects in natural scenes are typically larger and dominant). Remote sensing datasets are characterized by high proportions of small objects. For instance, 99.6% of objects in the DOTA dataset occupy less than 1% of the image area, significantly higher than the 6.6% observed in the VOC dataset. Utilizing only background features achieves a detection accuracy comparable to using only foreground features [23], suggesting that detection models have at least two learning modes. Therefore, the optimal solutions for teacher and student models may diverge in object detection in remote sensing images with pronounced spatial heterogeneity. Applying the same uniform feature distillation method to foreground and background regions prevents the student model's ability to converge to the global optimum.
2) Intermediate features in remote sensing object detection exhibit significant multitask decoupling divergence in the semantic dimension. Unlike image classification, which utilizes semantically consistent features for a single global task, object detection necessitates the concurrent construction of localization-sensitive spatial representations and classification-sensitive semantic representations within a shared feature space. Local and global spatial representations in the same feature space. This conflict is especially pronounced in remote sensing scenarios. Spatial heterogeneity results in complex background interference in classification. Thus, discriminative high-level semantic features are required (as illustrated in Fig. 1, where the backgrounds are highly diverse and dominant). Local tasks require comprehensive context modeling to capture geometrically informative low-level features. Furthermore, features extracted by shared convolutional layers can cause inter-task semantic conflicts, i.e., the progressive abstraction in higher layers reduces the fine-scale spatial information essential for precise localization, while lower level features lack robust discriminative semantic encoding [24], [25], [26].
3) Remote sensing object detection depends significantly on multiscale features [26], [27] due to its inherent scene characteristics: extreme variations in object scale (as shown in Fig. 1, with targets covering tens of pixels, such as ships, coexisting with targets covering hundreds of pixels, like ports) and complex backgrounds (Fig. 1 illustrates diverse backgrounds, including ports, suburban, and urban areas). Multiscale features integrate local details with global context, enhancing detection accuracy in complex scenes. However, divergent representational capacities in the solution spaces of teacher and student models lead to critical information loss using traditional single-scale distillation, adversely impacting target recognition and localization precision.

To address these challenges, we propose a nonuniform KD based on a feature decoupling (NKDFD) framework. In this work, "nonuniform" refers to a processing mechanism that is spatially asymmetric with respect to foreground and background features. Its core innovation is the decoupling of the foreground and background features in the representation space and performing independent nonuniform feature adaptation for different attributes. The framework generates 2-D Gaussian spatial masks from annotations to decouple teacher–student features into foreground and background components. Theoretically, these Gaussian masks help to decouple foreground and background features, thereby alleviating feature entanglement. Due to diverse detection patterns (i.e., foreground-only or background-only features), NKDFD implements independent feature adaptation, i.e., nonuniform feature extraction and knowledge transfer on the decoupled features. It enables the student model to identify optimal detection patterns in its solution space under the supervision of teacher features. Inspired by Li et al. [28], who employed Kolmogorov–Arnold networks (KANs) to construct low-parameter units for efficient high-dimensional semantic representation, we design a KAN-based semantic decoupling and selection (KSDS) module for nonuniform adaptation to deal with semantic discrepancies in high-dimensional feature spaces. This module performs semantic decoupling in the high-dimensional space, addressing the limitations of traditional methods in handling complex semantic entanglement. Building upon the NKDFD framework, we introduce a cross-scale feature interaction (CFI) mechanism that facilitates knowledge transfer across scales between teacher and student models separately for foreground and background features, improving the student's multiscale feature representation capabilities. In summary, our contributions are as follows.

1) An NKDFD framework is proposed to deal with spatial heterogeneity. Theoretically, the use of 2-D Gaussian priors allows for the spatial decoupling of foreground and background features. Experimental results show that the nonuniform distillation mechanism enhances detection robustness and localization accuracy in student models.
2) The KSDS and CFI modules are proposed to address semantic discrepancies and the critical need for multiscale feature representation in remote sensing object

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detection, while mitigating entropy reduction effects caused by the introduction of decoupling priors.

3) Empirical analysis confirms that the nonuniform knowledge transfer is necessary for KD in remote sensing object detection. If attribute-specific adaptation is not used, the output distribution is not statistically significantly different from that of random data.

The remainder of this article is organized as follows. Section II briefly reviews recent research in object detection and KD. Section III details the proposed methodology. Section IV presents the experimental results and discussion, including an investigation of the relationship between heterogeneous feature attributes (foreground/background) and their attribute-specific adapters. Section V concludes this study.

II. Related Works

A. Object Detection

Object detection, a pivotal task in intelligent image interpretation, precisely localizes and categorizes objects. Methods are classified by anchor dependency. Anchor-based approaches accelerate convergence via predefined anchor scales, whereas anchor-free methods predict object locations/categories from feature maps, offering superior adaptability. Single-stage detectors concurrently perform regression and classification for efficiency, whereas two-stage detectors generate region proposals before refinement, typically achieving higher accuracy. Remote sensing detection adapts these frameworks with enhancements for scale variations, aspect ratios, and orientations. Examples include Cao et al.'s rotation-invariant region-based convolutional neural network (R-CNN) for aircraft detection [39], Wang et al.'s dilated spatial pyramid for scale inconsistency [40], Xu et al.'s pseudo-anchor method for feature alignment [41], and Sun et al.'s contextual deformable modules for complex geometries [42]. Global to local enhances the feature representation of small objects and improves their detection accuracy through global semantic interaction and a local attention pyramid structure [43]. CAMCFormer addresses the limited receptive field of CNNs and feature alignment difficulties via global feature interaction and multidimensional relation modeling [44]. MLII mitigates feature misalignment caused by scale, orientation, and background variations in few-shot detection by means of semantic enhancement and feature cross-aggregation modules [45]. In the field of object detection, particularly for addressing challenges in complex environments, researchers have recently proposed various innovative methods. For infrared small target detection, RISTDnet integrates handcrafted features with convolutional networks, effectively enhancing detection robustness under low signal-to-noise ratio conditions [46], [47]. In multiobject tracking, to cope with severe occlusion, DCP-MOT [48] adopts a dynamic center point generation mechanism to strengthen target representation, while DD-MOT [49] utilizes fine-grained point trajectories to achieve more accurate trajectory recovery and association. Although these advances optimize architectures and feature representations, high-precision models have high computational costs and storage requirements that hinder deployment. It necessitates techniques that maximize model compression while preserving accuracy.

B. Knowledge Distillation

KD techniques are categorized into response-based, relation-based, and feature-based methods [50]. Feature distillation, which processes intermediate features from teacher and student models, has superior knowledge transfer efficiency and performance. However, when applied to complex tasks, such as object detection (particularly in remote sensing), its effectiveness is compromised by spatial heterogeneity, semantic discrepancies, and scale diversity. These challenges manifest as significant accuracy degradation during model lightweighting.

Most studies have applied different weights to different spatial locations during feature distillation by transferring spatial priors derived from ground truth or teacher models to student models directly or indirectly. This strategy is referred to as nonuniform KD in this work and is performed to address spatial heterogeneity.

In contrast, uniform distillation frameworks do not apply additional spatial weighting to the features of the teacher and student models, as illustrated in Fig. 2(a). These methods typically use evaluation metrics, including centered kernel alignment (CKA) [32], Pearson correlation (PKD) [21], the $L2$ distance (FT [29] and ReviewKD [31]), to measure the consistency between intermediate features. These metrics quantitatively assess the student's ability to learn the feature representations from the teacher model, thereby enabling knowledge transfer. After a comprehensive investigation, we categorized nonuniform feature distillation methods into three representative frameworks, as illustrated in Fig. 2(b)–(d).

First, as shown in Fig. 2(b), nonuniform KD based on a loss function improves knowledge transfer by employing spatially differentiated loss balancing methods that dynamically assign weights to foreground and background regions. Representative implementations like ACFG [51] and MasKD [34] apply adaptive weights to critical features, suppressing noise propagation.

Second, Fig. 2(c) illustrates an implicit feature decoupling strategy for nonuniform feature distillation between teacher and student models, using teacher model attention maps as guidance. For instance, Zagoruyko and Komodakis [35] ensured that student models mimicked teachers' feature attention patterns, enhancing feature representation. Similarly, AFD [52] generates spatial and channel attention maps from pooled intermediate features to constrain student learning. This indirect approach improves the student model's ability to capture information by acquiring the teacher's attention patterns.

Although these two categories of methods improve the performance of remote sensing object detection, they have significant limitations in knowledge transfer. The first achieves only coarse-grained feature balancing via loss weight adjustments, whereas the second exhibits insufficient decoupling due to indirect attention map guidance. For example, FT [29] introduces a feature adaptation module between the teacher and student features as a transition step, instead of directly

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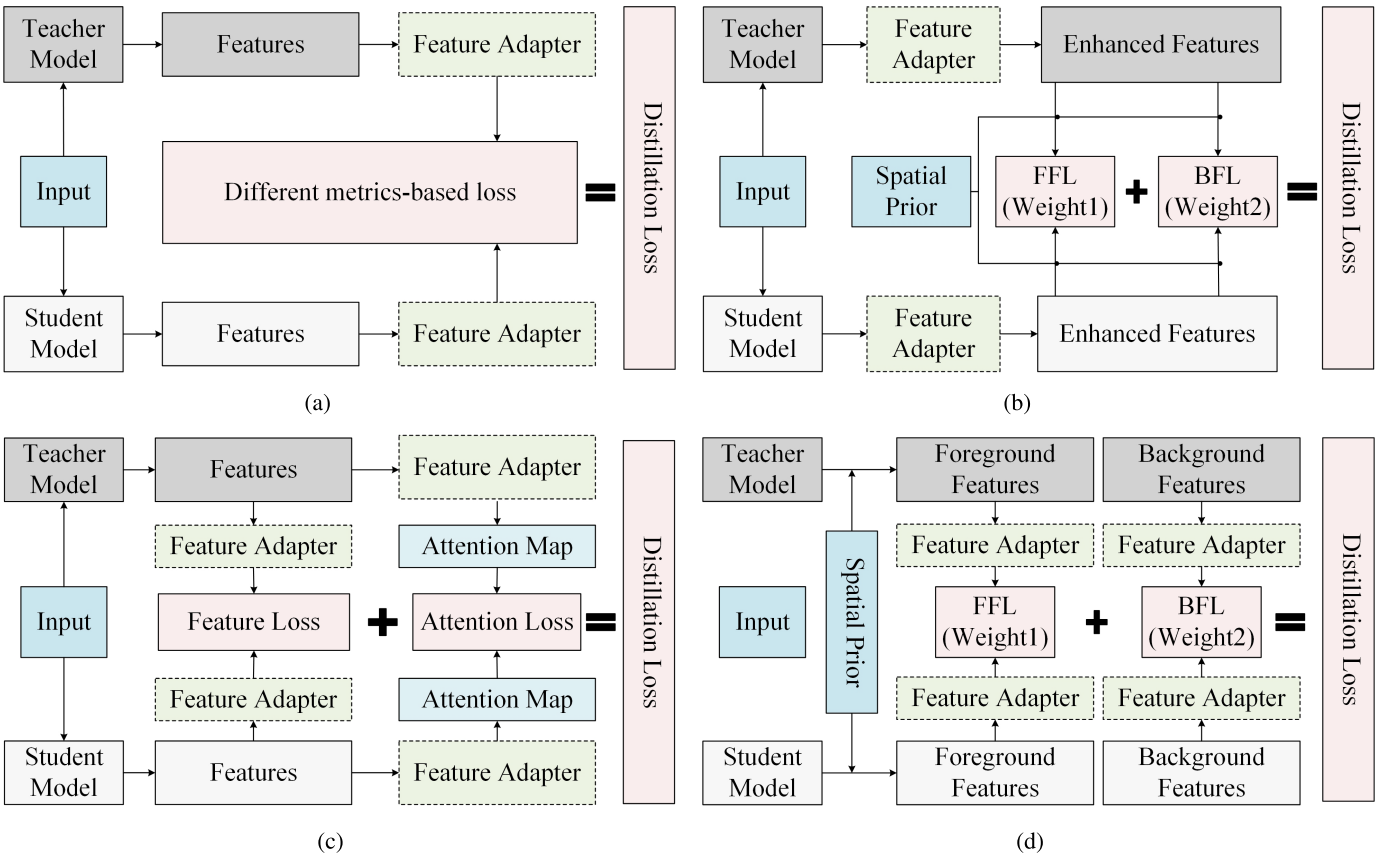


Fig. 2. Characteristics of different feature distillation frameworks (FFL: foreground feature loss and BFL: background feature loss). (a) Uniform KD framework (FT [29], OFD [30], ReviewKD [31], CKA [32], DIST [33], and PKD [21]). (b) Nonuniform KD based on loss function design (MasKD [34] and MGD [22]). (c) Nonuniform KD based on implicit spatial decoupling (AT [35], AB [36], FGD [37], FBKD [38], and CWD [20]). (d) NKDFD (proposed framework in this article: NKDFD).

comparing the two. However, such methods typically treat the entire feature set as the transfer target, without fully considering the inherent attribute difference between foreground and background features in the spatial dimension. MGD [22] enhances the student model's ability to absorb teacher features by randomly generating spatial masks applied to the student's features during KD. While this idea falls under the category of nonuniform feature processing proposed in our work, it relies on randomly generated spatial masks without accounting for the distinct spatial attributes of foreground versus background features, leading to potential mixing of foreground and background characteristics. These constraints limit the ability to compress the models and the models' performance. In contrast, the proposed NKDFD [see Fig. 2(d)] has the following advantages: 1) it decouples features into foreground and background components; 2) it employs independent feature adapters for each; and 3) it computes a balanced distillation loss. This design mitigates mutual interference between attribute-specific features (e.g., foreground/background) during distillation and alleviates the inherent distillation bias caused by orders-of-magnitude discrepancies between foreground and background regions.

Existing KD approaches typically employ CNN/multilayer perceptron (MLP)-based attention mechanisms to ensure semantic diversity. However, conventional approaches have inherent limitations in modeling high-dimensional nonlinear feature distributions. Due to their inability to model nonlinear relationships, they require more parameters but often cannot describe complex correlations. Li et al. [28] addressed this by using a residual KAN-CAB module, achieving a low-parameter high-dimensional representation applicable to remote sensing object detection features.

Some distillation frameworks implement cross-stage feature distillation to enhance the students' ability to obtain multiscale features from teachers. ReviewKD [31] strengthens hierarchical feature comprehension by using cross-layer connections that facilitate multiscale information exchange. ARSD [10] constructs a multiscale core feature imitation module to achieve comparison between the teacher and student models at multiscale feature levels, thereby optimizing the student model's ability to represent multiscale features and improving its object detection performance.

III. Methods

The proposed NKDFD framework (see Fig. 3) incorporates Gaussian distribution-based spatial decoupling (GDSD) to achieve spatial decoupling of intermediate features between the teacher and student models, obtaining attribute-specific features (foreground and background). It uses the decoupled

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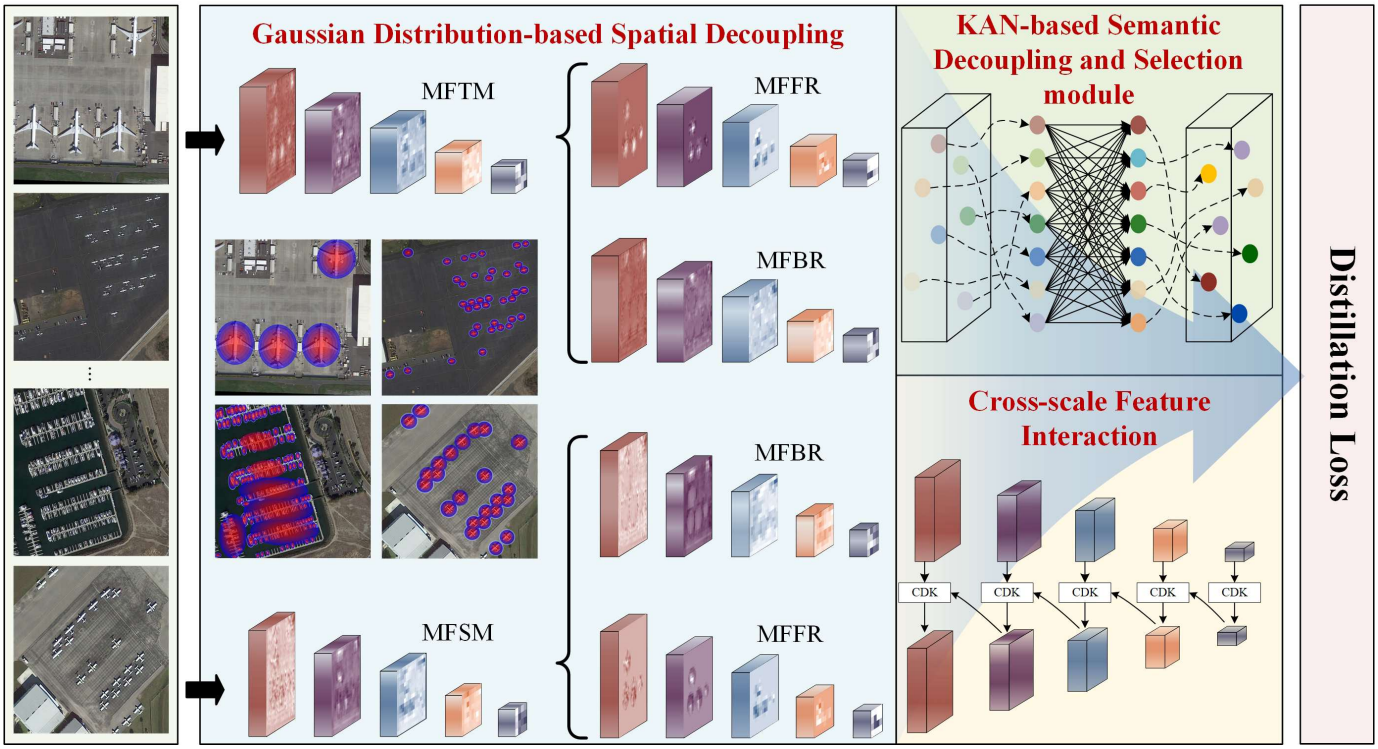


Fig. 3. Schematic of the proposed nonuniform KD framework (MFTM and MFSM denote the multiscale features of the teacher and student models, respectively; MFFR and MFBR represent the foreground and background multiscale features, respectively).

features to perform asymmetric feature adaptation and knowledge transfer, which is called nonuniform feature extraction and knowledge transfer in this article. The process uses two modules.

1) *KSDS:* It leverages the KAN's strong nonlinear modeling capability to decouple high-dimensional semantic spaces and adaptively extracts task-critical features.
2) *CFI:* It performs cross-scale knowledge transfer between teacher and student models, overcoming the limitations of single-scale feature mapping constraints. This strategy improves the student model's multiscale representation capability, enabling architecture-adapted feature construction.

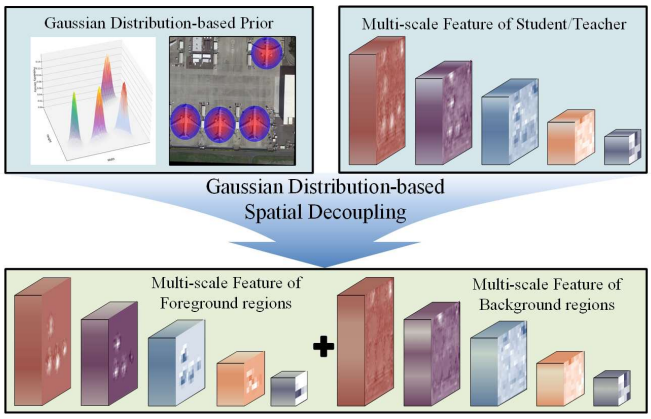


Fig. 4. Illustration of the GDSD mechanism.

A. Gaussian Distribution-Based Spatial Decoupling

The NKDFD framework's operation depends on the spatial decoupling of foreground/background features between teacher and student models. This module uses 2-D Gaussian masks and incorporates spatial heterogeneity priors, which are applied to intermediate features of both models to isolate region-specific attributes and reduce feature interference. Scale-dependent decoupling is used for remote sensing targets by tightly coupling object dimensions (w, h) with Gaussian distribution parameters (Σ). Fig. 4 illustrates this module's architecture.

1) Feature Spatial Decoupling Based on Gaussian Distribution: In remote sensing object detection, each instance is annotated with spatial location labels. For instance, the coordinates can be represented as "x_c, y_c, w, and h," where x_c and y_c denote the center coordinates of the rectangular box and w and h represent its width and height, respectively.

The probability density function of a 2-D Gaussian distribution is defined as

$$f(x,y) = \frac{1}{2\pi|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(p-\boldsymbol{\mu})^{\mathrm{T}}\Sigma^{-1}(p-\boldsymbol{\mu})\right) \quad (1)$$

where $p = [x, y]$ denotes a spatial coordinate vector, $\boldsymbol{\mu} = [x_c, y_c]$ represents the mean vector determined by the center coordinates of the bounding box, and Σ is the covariance matrix associated with the height h and width w of the bounding box, which is expressed as

$$\Sigma = \begin{bmatrix} \sigma_x^2 & 0 \\ 0 & \sigma_y^2 \end{bmatrix} = \begin{bmatrix} \left(\frac{w}{2}\right)^2 & 0 \\ 0 & \left(\frac{h}{2}\right)^2 \end{bmatrix}. \quad (2)$$

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The function $f(x, y)$ can be discretized to create a 2-D Gaussian mask $\widetilde{f}_{\text{frg_mask}}$ for the object foreground region. The background mask is defined as $\widetilde{f}_{\text{bkg_mask}} = 1 - \widetilde{f}_{\text{frg_mask}}$. Notably, the foreground and background masks exhibit mutual orthogonality or near orthogonality. To quantify this relationship, we calculate the inner product of the matrix between the two masks as follows:

$$\left\langle \widetilde{f}_{\text{frg_mask}}, \widetilde{f}_{\text{bkg_mask}} \right\rangle = \sum \left(\widetilde{f}_{\text{frg_mask}} - \widetilde{f}^2_{\text{frg_mask}} \right). \tag{3}$$

When $\widetilde{f}_{\text{frg_mask}}$ is a binary mask (0 or 1), it satisfies $\widetilde{f}_{\text{frg_mask}} = \widetilde{f}^2_{\text{frg_mask}}$. The inner product $\langle \widetilde{f}_{\text{frg_mask}}, \widetilde{f}_{\text{bkg_mask}} \rangle$ equals 0. This orthogonality implies reduced coupling in the solution space because the foreground and background masks are spatially disjoint.

When $\widetilde{f}_{\text{frg_mask}}$ is a Gaussian mask with $\sum \widetilde{f}_{\text{frg_mask}} = 1$, the inner product of the normalized matrix (matrix cosine similarity) is

$$\cos\theta = \frac{\langle \widetilde{f}_{\text{frg_mask}}, \widetilde{f}_{\text{bkg_mask}} \rangle}{\left\| \widetilde{f}_{\text{frg_mask}} \right\| \cdot \left\| \widetilde{f}_{\text{bkg_mask}} \right\|}. \tag{4}$$

The squared norms of the foreground and background masks, which are denoted as $\|\widetilde{f}_{\text{frg_mask}}\|$ and $\|\widetilde{f}_{\text{bkg_mask}}\|$, and the inner product of the matrix are computed as follows:

$$\left\| \widetilde{f}_{\text{frg_mask}} \right\| \approx \sqrt{\int f(x,y)^2 \, dxdy} = \sqrt{\frac{1}{4\pi\sigma_x\sigma_y}} \tag{5}$$

$$\begin{aligned} \left\| \widetilde{f}_{\text{bkg_mask}} \right\| &= \sqrt{\sum_{i,j} \left(1 - \widetilde{f}_{\text{frg_mask}}\right)^2} \\ &= \sqrt{n^2 - 2 + \frac{1}{4\pi\sigma_x\sigma_y}} \end{aligned} \tag{6}$$

$$\left\langle \widetilde{f}_{\text{frg_mask}}, \widetilde{f}_{\text{bkg_mask}} \right\rangle \approx 1 - \frac{1}{4\pi\sigma_x\sigma_y} \tag{7}$$

where n denotes the height/width of the mask and σ_x and σ_y denote the standard deviations in the x- and y-directions, respectively. By combining (4)–(7), the cosine of the angle between the foreground and background masks is derived as follows:

$$\cos\theta \approx \left(1 - \frac{1}{4\pi\sigma_x\sigma_y}\right) \frac{\sqrt{4\pi\sigma_x\sigma_y}}{n}. \tag{8}$$

For sufficiently large n values, the cosine similarity between the matrices approximates 0, indicating near orthogonality between foreground and background masks. It reduces the extent of spatial coupling. For the entire image, we construct multiple such Gaussian distributions (one per target) and at each spatial location (x, y), take the maximum value among all K Gaussian distributions, as shown in the following equation:

$$F_{\text{frg}}(x,y) = \max\left[f_1(x,y), f_2(x,y), \ldots, f_K(x,y) \right]. \tag{9}$$

The overall background mask is correspondingly defined as $F_{\text{bkg}} = 1 - F_{\text{frg}}$, where 1 represents an all-ones matrix of the same size as the image. It should be noted that remote sensing images typically exhibit a bird's-eye view perspective with almost no overlap between targets. Moreover, by applying the sigma principle, we constrain the foreground regions to a relatively limited spatial extent, and the nondominant regions of different targets have negligible influence on the dominant regions of other targets. Consequently, the following approximation can be made.

1) The total energy of the foreground mask is approximately equal to the sum of the energy of all individual target masks

$$\sum_{(x,y)} F_{\text{frg}}(x,y) \approx \sum_{k=1}^{K} \sum_{(x,y)} f_k(x,y) = K. \tag{10}$$

2) The squared sum of the foreground mask can be expressed as

$$\sum_{(x,y)} F^2_{\text{frg}}(x,y) = \sum_{(x,y)\in R_0} F^2_{\text{frg}}(x,y) + \sum_{k=1}^{K} \sum_{(x,y)\in R_k} F^2_{\text{frg}}(x,y) \tag{11}$$

where R_0 denotes the background region and R_k represents the foreground region of the kth target. Within the background region, the Gaussian responses of all targets approach 0: $\sum_{(x,y)\in R_0} F^2_{\text{frg}}(x,y) \approx 0$.

In the target-dominated regions, the maximum value operation ensures that the mask value at each location primarily stems from the Gaussian response of the corresponding target. Under the bird's-eye perspective typical of remote sensing imagery, overlapping contributions from other targets within a given target's dominant region have negligible influence

$$\sum_{(x,y)\in R_k} F^2_{\text{frg}}(x,y) \approx \sum_{(x,y)} f^2_k(x,y) = \frac{1}{4\pi \cdot \sigma_{x,k} \cdot \sigma_{y,k}} \tag{12}$$

$$< F_{\text{frg}}, F_{\text{bkg}} > \approx K - \sum_{k=1}^{K} \frac{1}{4\pi \cdot \sigma_{x,k} \cdot \sigma_{y,k}}. \tag{13}$$

Meanwhile, the $L2$ norm of the foreground mask and background mask is given by

$$\left\| F_{\text{frg}} \right\| = \sqrt{\sum_{(x,y)} F^2_{\text{frg}}(x,y)} \approx \sqrt{\sum_{k=1}^{K} \frac{1}{4\pi \cdot \sigma_{x,k} \cdot \sigma_{y,k}}} \tag{14}$$

$$\begin{aligned} \left\| F_{\text{bkg}} \right\| &= \left(n^2 - 2\sum_{(x,y)} F_{\text{frg}}(x,y) + \sum_{(x,y)} F^2_{\text{frg}}(x,y) \right)^{1/2} \\ &= \left(n^2 - 2K + \sum_{k=1}^{K} \frac{1}{4\pi \cdot \sigma_{x,k} \cdot \sigma_{y,k}} \right)^{1/2}. \end{aligned} \tag{15}$$

Since the image size n is typically significantly larger than both $2K$ and $\sum_{k=1}^{K} 1/(4\pi \cdot \sigma_{x,k} \cdot \sigma_{y,k})$, so we have $\|F_{\text{bkg}}\| \approx n$. Finally, the cosine similarity between the foreground and background masks can be expressed as

$$\cos\theta = \frac{< F_{\text{frg}}, F_{\text{bkg}} >}{\left\| F_{\text{frg}} \right\| \cdot \left\| F_{\text{bkg}} \right\|} \approx \frac{K - \sum_{k=1}^{K} \frac{1}{4\pi \cdot \sigma_{x,k} \cdot \sigma_{y,k}}}{n \cdot \sqrt{\sum_{k=1}^{K} \frac{1}{4\pi \cdot \sigma_{x,k} \cdot \sigma_{y,k}}}}. \tag{16}$$

Theoretically, as n increases, this value approaches 0, achieving an effect similar to that of binary hard decoupling. Consequently, under the reasonable assumption of minimal target overlap in remote sensing images, the foreground masks constructed based on 2-D Gaussian distributions can

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effectively separate features of different attributes and reduce feature coupling, even when extended to multitarget scenarios. This approach supports independent processing of foreground and background features during feature adaptation while avoiding issues such as feature discontinuity caused by complete hard decoupling.

2) Scale-Adaptive Adjustment: Multiscale objects in remote sensing images present feature extraction challenges. Although large objects contain sufficient foreground information, the background noise in the bounding boxes is substantial, interfering with foreground feature discrimination. Conversely, it is challenging to obtain sufficient features from small objects due to their small area. To address scale-dependent limitations in feature representation, we propose a scale-adaptive adjustment that dynamically parameterizes the spatial extent of 2-D Gaussian kernels according to the object's size. According to the definition of small, medium, and large targets in the Common Objects in Context (COCO) dataset [53] and their relative area ratios, we dynamically adjust the range of 2-D Gaussian distributions using the 3-sigma principle. The adaptive parameterization is expressed as

$$F_{\text{score}} = \int_{\mu-m\sigma}^{\mu+m\sigma} f(x,y)\,dx\,dy = \begin{cases} 0.6826, & m=1 \\ 0.9544, & m=2 \\ 0.9974, & m=3. \end{cases} \tag{17}$$

The range of the 2-D Gaussian distribution is determined by coupling the object scale with the parameter m using the sigma principle and employing the Mahalanobis distance

$$D^2 = z^{\mathrm{T}}\Sigma^{-1}z \tag{18}$$

where $z = \begin{bmatrix} x-\mu_x \\ y-\mu_y \end{bmatrix} = \begin{bmatrix} x-x_c \\ y-y_c \end{bmatrix}$ denotes the deviation vector from the point (x,y) to the mean vector $\boldsymbol{\mu} = [x_c, y_c]$. When $D^2 \le m^2$, the point (x,y) is considered within the $m\sigma$ range, where the value of m is determined by the area ratio of the object's ground-truth bounding box relative to the image

$$m = \begin{cases} 3, & \text{if area_ratio} \le 0.001, \\ 2, & \text{if } 0.001 < \text{area_ratio} \le 0.1 \\ 1, & \text{if area_ratio} > 0.1. \end{cases} \tag{19}$$

Based on the above equations, the 2-D Gaussian distribution for objects with varying area ratios can be derived as

$$f'(x,y) = \begin{cases} f(x,y), & \text{if } D^2 \le m^2 \\ 0, & \text{otherwise.} \end{cases} \tag{20}$$

The proposed scale-adaptive coupling mechanism establishes a relationship between the target dimensions and the 2-D Gaussian ranges to enable spatial feature decoupling in remote sensing detectors using Gaussian masking. This approach incorporates spatial heterogeneity priors to adjust loss weights dynamically, addressing the foreground–background imbalance while enabling independent feature distillation via strong decoupling. The method reduces solution space complexity and accelerates convergence for lightweight detection networks by removing tightly coupled optimization constraints.

3) Feature Adaptation: Feature adaptation typically involves feature extraction and knowledge transfer. Current remote sensing object detection methods employ feature distillation via adaptation [see (21) and (22)], uniformly adapting foreground and background regions for homogeneous distillation. However, significant magnitude discrepancies exist between foreground and background features (e.g., 1% foreground versus 99% background, as shown in Fig. 1). This imbalance leads to attribute feature interference, preventing the student model's optimization in the solution space

$$\overline{F}_s = H_s(F_s) \tag{21}$$

$$\overline{F}_t = H_t(F_t). \tag{22}$$

This article proposes spatial decoupling of intermediate features in teacher and student models using a 2-D Gaussian mask, followed by independent feature adaptation to achieve nonuniform distillation. This approach has two issues: mutual interference between foreground and background features and attribute bias caused by their magnitude discrepancies. The decoupling strategy enhances the student's feature representation, allowing adaptive approximation of task patterns without bias toward dominant distributions. Let F_t and F_s be the teacher and student features, respectively. The independent adaptation is expressed as

$$F_s^{\text{frg}} = H_s^{\text{frg}}\left(\widetilde{f}_{\text{frg}} \cdot F_s\right) \tag{23}$$

$$F_s^{\text{bkg}} = H_s^{\text{bkg}}\left(\widetilde{f}_{\text{bkg}} \cdot F_s\right) \tag{24}$$

$$F_t^{\text{frg}} = H_t^{\text{frg}}\left(\widetilde{f}_{\text{frg}} \cdot F_t\right) \tag{25}$$

$$F_t^{\text{bkg}} = H_t^{\text{bkg}}\left(\widetilde{f}_{\text{bkg}} \cdot F_t\right) \tag{26}$$

where the variables H_s^{frg} and H_s^{bkg} denote the feature adapters designed for foreground and background regions in the student model, respectively. Similarly, H_t^{frg} and H_t^{bkg} represent the feature adapters for foreground and background features in the teacher model. These adapter designs are described in Sections III-B and III-C. The terms F_s^{frg} and F_s^{bkg} represent the student model's foreground and background features after processing using the adapters. Analogously, F_t^{frg} and F_t^{bkg} denote the teacher model's foreground and background features after processing using the feature adapters.

B. KSDS Module

We propose the KSDS module to address the semantic discrepancy of intermediate features. It improves the student model's nonlinear modeling capacity to extract semantic information from different attributes adaptively.

1) KSDS Module: The proposed KSDS module (depicted in Fig. 5) utilizes two feature maps of different sizes as inputs. The smaller feature map is interpolated to align its size with another feature map. The larger feature map is processed using convolutional layers for feature extraction, and the results are concatenated with the interpolated feature map. Subsequently, the concatenated feature map undergoes adaptive average pooling, followed by KAN processing to enable nonlinear modeling in the high-dimensional space, achieving semantic

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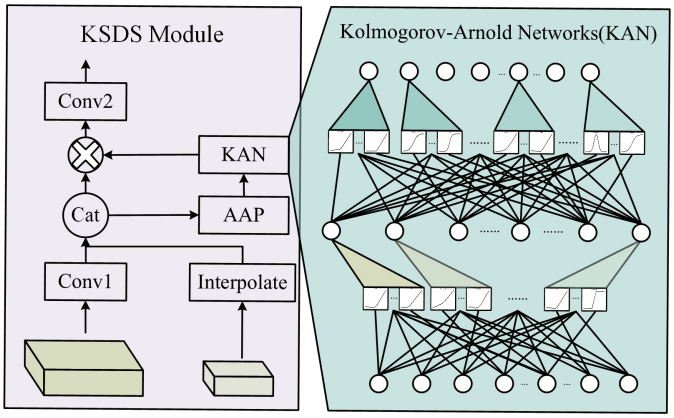


Fig. 5. Illustration of the KSDS module (Conv1 and Conv2: convolutional layers, AAP: adaptive average pooling, and interpolate: interpolation).

decoupling. The final output is generated using an additional convolutional layer

$$F_{\text{in}} = \text{Cat}\left(\text{Conv}_1(x), \text{Interpolate}(y)\right)$$
$$F_{\text{out}} = \text{Conv}_2\left(F_{\text{in}} \cdot \text{KAN}\left(\text{Adaptative_avg_pool}(F_{\text{in}})\right)\right) \quad (27)$$

where Interpolate denotes the interpolation to align the spatial scale, Conv_1 and Conv_2 represent the two convolutional modules, Cat indicates the feature concatenation along the channel dimension, Adaptative_avg_pool refers to the global adaptive average pooling, which adjusts the output spatial dimensions dynamically, KAN stands for the Kolmogorov–Arnold network, a module for nonlinear modeling (as defined in this article), and x and y denote the multiscale feature inputs from the current layer (larger feature map) and previous layer (smaller feature map), respectively.

Based on the spatial decoupling of foreground/background features via 2-D Gaussian distributions, we construct attribute-specific feature adapters using the proposed KSDS module to enable nonuniform KD. These adapters correspond to the decoupled features H_s^{frg}, H_s^{bkg}, H_t^{frg}, and H_t^{bkg} defined above.

2) Semantic Decoupling: Existing solutions for dealing with semantic discrepancies in intermediate features in remote sensing object detection involve channel attention mechanisms based on CNN/MLP or Transformer. The processing logic of CNN/MLP for input f is shown in (28), where w_i represent the linear weights and f_i corresponds to the feature associated with the ith semantic dimension. It involves performing a weighted summation of the input features, followed by a nonlinear activation to obtain the output features, with the nonlinear capability provided by the σ function:

$$y_i' = \sigma(w_1 f_1 + w_2 f_2 + w_3 f_3 + w_4 f_4 + \ldots). \quad (28)$$

The mapping of a single-layer Transformer can be formalized by (29), where b_i represent the linear bias. Attn(f) can be regarded as the weighted average of the input vector f, and the nonlinear activation is applied to Attn(f) through the function σ. That is, its nonlinear capability is likewise provided solely by the σ function

$$\begin{cases} \text{Attn}(f) = \text{soft max}\left(QK^{\text{T}} / \sqrt{d}\right) \cdot V \\ y_i' = w_1 \cdot \sigma\left[w_2 \cdot \text{Attn}(f) + b_1\right] + b_2. \end{cases} \quad (29)$$

In contrast, KAN directly constructs multivariate functions based on the Kolmogorov–Arnold representation theorem, as shown in (30). Here, the "weight" on each edge (denoted as w_i or b_i) is replaced by a learnable 1-D function φ_i, making each connection itself a nonlinear mapping. This implies that KAN's nonlinear capability is structurally more fine-grained and parameter-efficient, enabling it to approximate more complex function families with fewer parameters

$$y_i'' = w_1 \cdot \varphi_1(f_1) + w_2 \cdot \varphi_2(f_2) + w_3 \cdot \varphi_3(f_3) + \ldots \quad (30)$$

Based on (28)–(30), we compare the nonlinear capabilities of the fundamental units in CNN/MLP, Transformer, and KAN as follows: the CNN/MLP unit performs mapping through a linear combination of inputs followed by a fixed nonlinear activation function, meaning its nonlinear expressive capability relies entirely on the predefined activation function. The Transformer unit consists of two stages: first, a weighted summation of inputs, followed by a nonlinear transformation via the softmax function, and then another nonlinear activation. Its overall nonlinearity is provided by two fixed nonlinear functions. In contrast, the KAN unit applies a learnable nonlinear transformation to each input separately before performing a weighted summation. Its nonlinear capability stems from multiple adaptively adjustable nonlinear functions, the number of which equals the input dimension. Theoretical analysis indicates that the KAN unit possesses stronger nonlinear expressive power, thereby exhibiting greater potential for feature decoupling in high-dimensional semantic spaces and making it more suitable for handling complex nonlinear feature interaction tasks.

C. CFI Module

We propose the CFI to address scale variability in remote sensing object detection, which requires multiscale features. We extend the conventional single-scale to single-scale decoupling distillation. This approach improves the solution space of the student model using multiscale features.

1) Cross-Scale Feature Interaction: The proposed CFI module (see Fig. 6) performs CFI through multiscale feature decoupling. It is lightweight and enables the student model to integrate multiscale feature information adaptively while efficiently leveraging prior knowledge derived from the teacher. Given an input image Img, let S and T be intermediate feature extractors for student and teacher models, respectively. The multiscale outputs are expressed as

$$(F_1^s, F_2^s, F_3^s, F_4^s, F_5^s) = S(\text{Img}) \quad (31)$$
$$(F_1^t, F_2^t, F_3^t, F_4^t, F_5^t) = T(\text{Img}) \quad (32)$$

where F_i^s and F_j^t denote the ith and jth scale features extracted from the intermediate layers of the student and teacher models, respectively.

The spatially decoupled foreground/background features are used, and the module progressively integrates multiscale features from the student model. Since the adapters for different attributes are not shared, foreground and background

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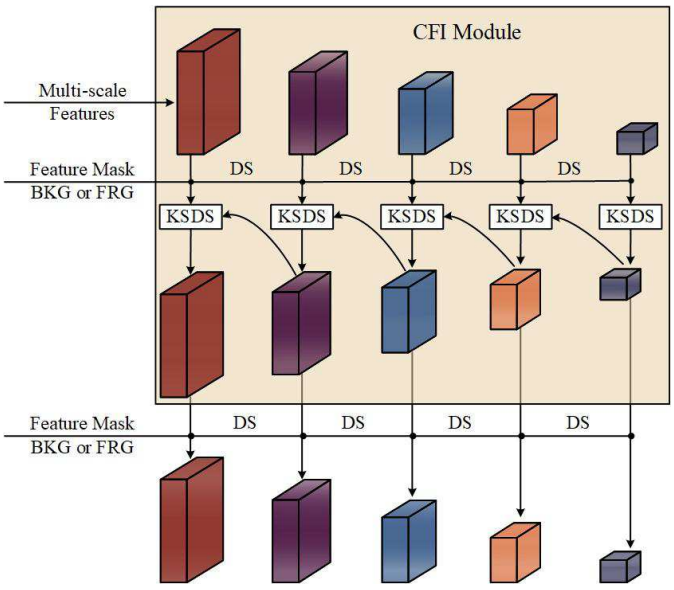


Fig. 6. Illustration of the CFI module (BKG: background, FRG: foreground, and DS: downsampling).

features undergo independent multiscale interactions, resulting in nonuniform feature integration

$$F^x_{k,\text{frg}} = \widetilde{f}_{\text{frg}} \cdot F^x_k \tag{33}$$

$$F^x_{k,\text{bkg}} = \widetilde{f}_{\text{bkg}} \cdot F^x_k \tag{34}$$

$$\overline{F}^x_{i,\text{frg}} = H^{\text{frg}}_x \left(F^x_{i,\text{frg}}, F^x_{i-1,\text{frg}}\right) \tag{35}$$

$$\overline{F}^x_{i,\text{bkg}} = H^{\text{bkg}}_x \left(F^x_{i,\text{bkg}}, F^s_{i-1,\text{bkg}}\right) \tag{36}$$

where H^{frg}_x and H^{bkg}_x denote the feature adapters of the KSDS module and $\overline{F}^x_{i,\text{frg}}$ and $\overline{F}^x_{i,\text{bkg}}$ represent the adapted foreground and background features, respectively. The subscript $x \in \{s, t\}$ indicates feature provenance (t: teacher model, s: student model), and $k \in \{1, 2, 3, 4, 5\}$ refers to the five feature scales.

2) Loss Function: The distillation loss is defined as follows:

$$\text{loss}_{\text{distill}} = \alpha \cdot \sum_{k=1}^{5} \text{loss}^{\text{frg}}_{\text{distill}} \left(\bar{F}^s_{k,\text{frg}}, \bar{F}^t_{k,\text{frg}}\right) + (1-\alpha) \cdot \sum_{k=1}^{5} \text{loss}^{\text{bkg}}_{\text{distill}} \left(\bar{F}^s_{k,\text{bkg}}, \bar{F}^t_{k,\text{bkg}}\right). \tag{37}$$

The above equation provides an abstract representation of the balancing mechanism between foreground and background losses. In practical experiments, the α factor adopts an adaptive adjustment mechanism to coordinate the proportional relationship between the two loss components. This design is motivated by the characteristics of remote sensing object detection scenarios, where foreground targets typically occupy a small proportion of the image, whereas background regions dominate. If the losses were simply summed without weighting, the substantially larger contribution from the background regions could dilute the knowledge transfer effect of the foreground features. Therefore, the adaptive weight α is introduced to balance the two components. Specifically, the value of α is dynamically obtained based on the foreground mask derived from the 2-D Gaussian distribution, allowing it to adapt to varying foreground–background ratios during KD.

IV. Experiments

This section describes the experiment performed with the proposed NKDFD framework. We introduce the datasets and parameter settings, present the experimental results and analysis, and discuss the findings.

A. Datasets

Experiments were conducted on remote sensing object detection datasets, including DOTA, DIOR, RSOD, and VHR10, covering objects of diverse categories and sizes in different detection scenarios.

DOTA is a large-scale benchmark dataset for object detection in aerial remote sensing images. It contains images from diverse sensors. The resolution ranges from 800 × 800 to 20 000 × 20 000 pixels, encompassing objects of diverse scales and shapes. It comprises 2806 remote sensing images and 188 282 object instances, covering 15 object categories: plane, ship, storage tank, baseball diamond, tennis court, basketball court, track field, harbor, bridge, large vehicle, small vehicle, helicopter, roundabout, soccer ball field, and swimming pool.

DIOR is a large-scale dataset for object detection in optical remote sensing images. It comprises 23 464 remote sensing images and 192 518 object instances and includes 20 object categories: airplane, airport, baseball field, basketball court, bridge, chimney, dam, expressway service area, expressway toll station, golf course, track field, harbor, overpass, ship, stadium, storage tank, tennis court, train station, vehicle, and windmill.

RSOD is a publicly available dataset for remote sensing object detection. It contains four categories: aircraft, oil tank, playground, and overpass. It comprises 976 images and 6950 object instances.

VHR10 is used for object detection in high-resolution remote sensing images and contains ten classes: airplane, ship, storage tank, baseball diamond, tennis court, basketball court, track field, harbor, bridge, and vehicle. It comprises 800 images, including 650 positive samples (containing objects) and 150 negative samples (no objects).

The DIOR, RSOD, and VHR10 datasets were split into training/testing sets using an 8:2 ratio. The images in the DOTA dataset were cropped to 1024 × 1024 pixels with a 400-pixel overlap, resulting in 24 111 training and 6028 testing images.

B. Implementation Details

The detection models include single-stage and two-stage algorithms, including anchor-based and anchor-free methods. We evaluated ATSS [54], GFL [55], VFNet [56], PAA [57], FCOS [58], RetinaNet [59], CenterNet [60], Tood [61], YOLOv5,[1] YOLOF [62], and YOLOX [63]. Two backbone networks were used: ResNet [64] and CSPDarknet [65]. YOLOF uses only single-scale features for classification and

[1]https://github.com/ultralytics/yolov5

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TABLE I
EVALUATION RESULTS FOR THE NKDFD MODEL'S GENERALIZATION PERFORMANCE ON THE DOTA DATASET

Class	Sub-Class	Method	mAP (%)	mAP_{50} (%)	mAP_{75} (%)	mAP_s (%)	mAP_m (%)	mAP_l (%)	$Params$ (M)	$FLOPs$ (G)
Single Stage Detector	Anchor-Based	RetinaNet(T)	43.3	66.5	45.8	6.2	41.8	67.1	55.384	64.590
		RetinaNet(S)	29.9	52.2	29.5	3.0	28.7	46.5	19.833	35.573
		RetinaNet(ours)	**37.7**	**60.6**	**40.4**	**3.9**	**35.5**	**56.6**	**19.833**	**35.573**
		PAA(T)	51.3	75.6	56.5	15.4	50.7	70.3	51.112	63.686
		PAA(S)	39.3	63.8	41.7	8.4	36.9	57.9	19.100	34.877
		PAA(ours)	**45.6**	**70.0**	**49.6**	**11.9**	**43.9**	**63.8**	**19.100**	**34.877**
		ATSS(T)	54.4	76.6	61.3	18.5	54.8	70.8	51.112	63.686
		ATSS(S)	43.6	67.7	47.7	9.6	43.9	58.1	19.100	34.877
		ATSS(ours)	**48.8**	**72.1**	**54.9**	**13.7**	**50.2**	**63.2**	**19.100**	**34.877**
		YOLOF(T)	21.7	40.0	20.6	1.9	21.4	35.5	61.6548	45.820
		YOLOF(S)	14.3	31.9	11.2	0.4	9.9	30.4	16.528	10.871
		YOLOF(ours)	**19.2**	**38.5**	**17.3**	**0.8**	**13.0**	**41.4**	**16.528**	**10.871**
		GFL(T)	53.9	75.9	60.9	17.9	54.8	70.4	51.283	64.521
		GFL(S)	49.5	71.4	55.6	13.7	50.8	64.3	19.271	35.712
		GFL(ours)	**51.4**	**72.8**	**57.8**	**15.2**	**53.3**	**65.8**	**19.271**	**35.712**
	Anchor-Free	CenterNet(T)	47.6	73.3	53.7	16.0	47.7	62.2	51.110	63.675
		CenterNet(S)	35.9	60.2	37.7	8.2	35.7	46.3	19.098	34.866
		CenterNet(ours)	**44.5**	**68.0**	**50.1**	**13.2**	**45.3**	**56.4**	**19.098**	**34.866**
		VFNet(T)	54.0	76.5	60.9	17.2	54.5	70.9	51.709	60.825
		VFNet(S)	44.0	67.9	48.6	10.4	44.2	57.9	19.697	32.017
		VFNet(ours)	**49.7**	**72.5**	**56.1**	**14.8**	**51.4**	**63.6**	**19.697**	**32.017**
		FCOS(T)	40.9	69.6	44.3	10.2	41.9	53.2	51.114	64.590
		FCOS(S)	39.1	66.3	41.5	9.2	40.1	50.8	19.103	35.573
		FCOS(ours)	**41.0**	**66.4**	**44.4**	**9.9**	**41.8**	**53.2**	**19.103**	**35.573**
		YOLOX(T)	60.7	84.1	71.9	35.8	65.4	66.5	54.159	49.720
		YOLOX(S)	32.4	58.7	34.2	11.3	34.5	36.9	5.037	3.203
		YOLOX(ours)	**33.1**	**60.2**	**34.9**	**11.5**	**35.9**	36.3	**5.037**	**3.203**
		TOOD(T)	56.4	77.9	64.0	20.2	56.1	73.1	51.017	62.745
		TOOD(S)	51.8	74.2	58.2	16.8	52.8	65.9	19.006	33.937
		TOOD(ours)	**52.6**	**74.4**	**59.6**	16.6	**53.4**	**66.8**	**19.006**	**33.937**
Two Stage Detector		FRCNN(T)	45.3	69.2	50.8	7.4	49.1	62.6	60.412	82.698
		FRCNN(S)	39.3	64.0	42.1	6.3	42.3	52.9	28.351	49.772
		FRCNN(ours)	**42.4**	**66.0**	**47.5**	**6.4**	**46.1**	**56.7**	**28.351**	**49.772**

localization. Consequently, not all the object detection models used multiscale features, introducing an additional experimental variable.

All experiments were conducted on Ubuntu 18.04.6 LTS using an NVIDIA RTX A5000 GPU with 24 256-MiB memory and an Intel Core i9-10900X CPU. Stochastic gradient descent (SGD) was employed as the optimizer for all student models and feature distillation to ensure fairness. The learning rate was 0.005, and the batch size was 16. The teacher models utilized ResNet101 or CSPDarknet-l as backbone networks, whereas the student models utilized ResNet18 or CSPDarknet-tiny architectures.

The evaluation metrics include mean average precision (mAP), number of parameters (Params), and number of floating-point operations (FLOPs). The mAP include several submetrics: mAP_{50}, mAP_{75}, mAP, mAP_s, mAP_m, and mAP_l. mAP_{50}, and mAP_{75} denote the average detection precision at intersection-over-union (IoU) thresholds of 0.5 and 0.75, respectively. mAP was calculated at IoU thresholds ranging from 0.5 to 0.95 in 0.05 increments. Following the COCO dataset, the mAP_s, mAP_m, and mAP_l correspond to the mAP for small, medium, and large objects, respectively. The Params and FLOPs are efficiency metrics, quantifying model storage requirements and computational resource consumption, respectively, with input images fixed at a resolution of 512 $\times$ 512 pixels.

C. Distillation Experiments

1) Distillation Results on DOTA: The 11 models were evaluated on the DOTA dataset (see Table I). The NKDFD framework improved student model performance across diverse architectures (single-stage, two-stage, anchor-based, and anchor-free). Anchor-based models (RetinaNet, PAA, ATSS, YOLOF, and GFL) showed mAP gains of 7.8%, 6.3%, 5.2%, 4.9%, 1.9%, 0.7%, and 0.7%, respectively.

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TABLE II
EVALUATION RESULTS FOR THE NKDFD MODEL'S GENERALIZATION PERFORMANCE ON THE DIOR DATASET

Class	Sub-Class	Method	mAP (%)	mAP_{50} (%)	mAP_{75} (%)	mAP_s (%)	mAP_m (%)	mAP_l (%)	$Params$ (M)	$FLOPs$ (G)
Single Stage Detector	Anchor-Based	PAA(T)	39.1	64.0	39.5	5.6	29.7	59.2	51.149	71.191
		PAA(S)	33.9	58.7	33.6	4.6	25.1	52.1	19.137	39.07
		PAA(ours)	**35.9**	**61.3**	**36.1**	**5.4**	**27.2**	**53.6**	**19.137**	**39.070**
		GFL(T)	39.5	63.0	40.5	6.8	31.0	59.0	51.294	71.983
		GFL(S)	34.7	57.7	34.9	5.2	26.3	53.1	19.283	39.862
		GFL(ours)	**36.0**	**59.3**	**36.9**	**5.5**	**28.3**	**53.9**	**19.283**	**39.862**
		YOLOF(T)	32.5	55.8	32.1	1.5	22.6	55.2	61.769	45.850
		YOLOF(S)	24.0	44.8	22.9	0.6	13.6	43.7	16.585	10.885
		YOLOF(ours)	**25.1**	**46.7**	**24.0**	0.4	**13.7**	**46.0**	**16.585**	**10.885**
		ATSS(T)	38.0	62.2	38.9	5.8	29.5	57.2	51.149	71.191
		ATSS(S)	34.3	58.4	34.3	5.1	26.3	51.6	19.137	39.070
		ATSS(ours)	**35.3**	**59.6**	**35.9**	**5.5**	**27.5**	**52.8**	**19.137**	**39.070**
		RetinaNet(T)	36.1	58.9	36.5	2.6	27.2	56.9	55.716	73.800
		RetinaNet(S)	30.9	53.3	30.5	2.0	22.5	50.1	20.165	41.453
		RetineNet(ours)	**31.7**	**54.7**	**31.7**	**2.2**	**23.5**	**50.2**	**20.165**	**41.453**
	Anchor-Free	FCOS(T)	34.6	60.9	33.9	5.1	26.5	52.8	51.151	71.191
		FCOS(S)	28.3	52.8	26.9	3.2	21.1	44.5	19.139	39.070
		FCOS(ours)	**30.2**	**55.6**	**28.9**	**4.2**	**23.7**	**46.1**	**19.139**	**39.070**
		TOOD(T)	39.9	62.7	41.2	7.5	32.0	59.4	51.054	70.143
		TOOD(S)	36.0	59.1	36.6	6.2	28.4	53.9	19.043	38.022
		TOOD(ours)	**37.6**	**61.1**	**38.5**	**6.8**	**29.5**	**55.6**	**19.043**	**38.022**
		YOLOX(T)	40.7	64.1	43.2	10.2	31.8	57.9	54.163	49.727
		YOLOX(S)	34.0	56.7	35.6	6.5	25.1	51.0	6.194	5.614
		YOLOX(ours)	**35.5**	**58.9**	**37.4**	**7.3**	**26.9**	**52.4**	**6.194**	**5.614**
		VFNet(T)	40.6	63.4	42.0	7.2	31.6	60.2	51.745	68.004
		VFNet(S)	36.1	59.8	36.7	5.4	27.9	54.2	19.734	35.883
		VFNet(ours)	**37.2**	**60.9**	**38.0**	**5.9**	**29.4**	**55.5**	**19.734**	**35.883**
Two Stage Detector		FRCNN(T)	38.5	61.3	40.4	5.3	31.1	58.7	60.438	82.724
		FRCNN(S)	35.6	58.9	37.0	4.7	28.0	54.4	28.377	49.797
		FRCNN(ours)	**36.2**	**60.0**	**37.6**	**5.2**	**28.4**	**54.9**	**28.377**	**49.797**

Anchor-free models (CenterNet, VFNet, Tood, YOLOX, and FCOS) achieved mAP improvements of 8.6%, 5.7%, 0.8%, and 0.9%, with CenterNet reaching teacher-level accuracy. Faster R-CNN (FRCNN, two-stage) exhibited a 3.1% improvement in mAP.

Notably, the improvements based on more stringent criteria (mAP_{75}: 1.4%–12.4% absolute and 2.4%–54.5% relative) exceeded those based on mAP_{50} (0.1%–7.8% and 0.2%–20.7%). For example, the improvement of CenterNet was 12.4% (32.9% relative) mAP_{75} versus 7.8% (13.0%) mAP_{50}, demonstrating high-quality detection. Furthermore, KD models like FCOS achieve teacher-comparable accuracy, using 1/3 of the Params and 1/2 of the FLOPs. The results based on the DOTA dataset are presented in Table I. They demonstrate that the NKDFD framework improved the performance of student models with diverse architectures, including single-stage, two-stage, anchor-based, and anchor-free models.

2) Distillation Results on DIOR: The results for ten models (see Table II) demonstrate that the NKDFD framework significantly improved the detection performance for diverse model types. Anchor-based models (PAA, GFL, YOLOF, ATSS, and RetinaNet) achieved mAP improvements of 2.0%, 1.3%, 1.1%, 1.0%, and 0.8%. Anchor-free models (FCOS, Tood, YOLOX, and VFNet) showed improvements of 1.9%, 1.6%, 1.5%, and 1.1%, and FRCNN (two-stage) exhibited a 0.6% mAP improvement.

The optimization efficiency was higher for more stringent criteria. Some models exhibited higher relative mAP_{75} improvements (3.5%–7.4%) compared with mAP_{50} (1.8%–5.3%). For example, PAA achieved a 2.5% absolute (7.4% relative) mAP_{75} improvement versus 2.6% (4.4% relative) for mAP_{50}. Similarly, GFL and ATSS showed 2.1× and 2.2× higher relative improvements in mAP_{75} (5.7% and 4.7%) than mAP_{50} (2.7% and 2.1%). FCOS exhibited a 7.4% relative mAP_{75} gain (1.4× its mAP_{50} improvement).

These results confirm that NKDFD substantially improved the prediction accuracy and robustness based on several evaluation metrics. KD models consistently showed improvements in mAP_{50} and mAP_{75}, as well as generalization and precision improvements.

3) Distillation Results on RSOD and VHR10: The proposed distillation framework was validated on RSOD

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TABLE III
EVALUATION RESULTS FOR THE NKDFD MODEL'S GENERALIZATION PERFORMANCE ON THE RSOD DATASET

Class	Sub-Class	Method	mAP (%)	mAP_{50} (%)	mAP_{75} (%)	mAP_s (%)	mAP_m (%)	mAP_l (%)	$Params$ (M)	$FLOPs$ (G)
Single Stage Detector	Anchor-based	GFL(T)	56.9	91.3	62.9	15.4	60.4	64.1	51.283	64.521
		GFL(S)	52.9	86.4	57.8	10.8	58.7	60.3	19.271	35.712
		GFL(ours)	**55.0**	**91.3**	**60.2**	10.6	**61.5**	**62.2**	**19.271**	**35.712**
		YOLOv5(T)	50.9	79.2	-	-	-	-	53.134	134.700
		YOLOv5(S)	43.9	71.5	-	-	-	-	2.504	7.100
		YOLOv5(ours)	**46.0**	**74.2**	-	-	-	-	**2.504**	**7.100**
		YOLOF(T)	41.2	78.7	38.1	2.0	31.1	53.4	61.401	41.037
		YOLOF(S)	24.9	59.5	17.3	0.3	15.4	35.4	16.401	9.720
		YOLOF(ours)	**26.6**	59.4	**19.1**	**0.9**	**18.1**	**35.7**	**16.401**	**9.720**
		PAA(T)	55.7	89.9	58.7	11.4	56.1	63.2	51.112	63.686
		PAA(S)	48.8	87.7	51.8	5.6	50.5	56.7	19.100	34.877
		PAA(ours))	**50.2**	**89.4**	**51.9**	**8.5**	49.7	**57.8**	**19.100**	**34.877**
		RetinaNet(T)	41.8	80.1	41.6	0.6	38.1	52.7	55.384	64.590
		RetinaNet(S)	35.8	76.3	34.0	0.3	29.8	47.2	19.833	35.573
		RetinaNet(ours)	**37.2**	**78.6**	32.7	**0.5**	**33.6**	47.0	**19.833**	**35.573**
		ATSS(T)	57.5	91.9	61.1	18.5	61.0	64.6	51.112	63.686
		ATSS(S)	52.6	87.4	59.7	13.3	58.2	60.0	19.100	34.877
		ATSS(ours)	**53.6**	**88.8**	57.5	12.1	58.2	**60.6**	**19.100**	**34.877**
	Anchor-free	CenterNet(T)	47.3	87.5	49.0	13.1	51.8	53.4	51.110	63.675
		CenterNet(S)	44.8	87.4	42.7	11.3	50.2	50.4	19.098	34.866
		CenterNet(ours)	**53.3**	**90.7**	**56.7**	**13.0**	**56.4**	**59.1**	**19.098**	**34.866**
		FCOS(T)	43.0	86.1	39.3	6.5	44.5	49.9	51.114	63.686
		FCOS(S)	39.7	81.4	35.0	5.1	42.0	46.0	19.103	34.877
		FCOS(ours)	**46.3**	**84.2**	**46.2**	**8.2**	**46.1**	**53.4**	**19.103**	**34.877**
		VFNet(T)	48.9	89.3	49.9	13.7	58.2	56.9	51.709	60.825
		VFNet(S)	42.8	87.0	37.4	9.4	54.0	51.1	19.697	32.017
		VFNet(ours)	**46.9**	**90.1**	**44.4**	7.6	53.0	**53.4**	**19.697**	**32.017**
		TOOD(T)	59.0	91.8	64.7	19.5	62.0	66.3	51.017	62.745
		TOOD(S)	54.5	86.3	59.3	14.7	60.7	61.6	19.006	33.937
		TOOD(ours)	**57.6**	**91.0**	**62.4**	**15.3**	**60.8**	**64.6**	**19.006**	**33.937**
		YOLOX(T)	50.5	91.5	49.9	27.0	66.2	57.2	54.150	44.579
		YOLOX(S)	44.3	80.1	45.9	19.7	58.0	49.8	5.034	3.200
		YOLOX(ours)	**46.1**	76.6	**52.0**	**19.6**	**57.3**	**52.4**	**5.034**	**3.200**
Two Stage Detector		FRCNN(T)	53.8	89.1	55.9	7.8	57.5	62.4	60.356	75.555
		FRCNN(S)	49.3	88.4	53.1	7.5	53.3	55.7	28.295	46.025
		FRCNN(ours)	**50.9**	**87.7**	52.9	**8.0**	**58.1**	**57.8**	**28.295**	**46.025**

(12 models) and VHR10 (ten models) (see Tables III and IV). On RSOD, anchor-based models (GFL, YOLOv5, YOLOF, PAA, RetinaNet, and ATSS) showed mAP gains of 1.0%–2.1%. Anchor-free models (CenterNet, FCOS, VFNet, Tood, and YOLOX) achieved greater improvements (3.1%–8.5%), with distilled CenterNet and FCOS surpassing teachers by 6.0% and 3.3% mAP and using 1/3 of the Params and 1/2 of the FLOPs. This result demonstrates student models exceeding teacher performance by using adaptive detection patterns. Furthermore, the mAP of the FRCNN student model increased from 49.3% to 50.9% on RSOD. On VHR10, the mAP saw a more substantial rise from 43.3% to 46.0%. These improvements were achieved without introducing any additional computational overhead during inference. It strongly validates the excellent generalizability of our distillation strategy even for two-stage detectors and further underscores the universality of our proposed method.

YOLOF showed significant gains on both datasets (+1.7% RSOD and +4.4% VHR10), demonstrating the GDSD and KSDS modules' synergy for single-scale models. The results confirm the framework's robustness across datasets.

Figs. 7–10 compare the Tood-R18 detection results before and after NKDFD distillation on DOTA, DIOR, RSOD, and VHR10. The visualizations show qualitative improvements. The red boxes indicate newly detected objects, and the yellow boxes show a reduction in false positives or duplicate detections. We used the same samples as in Fig. 1. The results indicate performance improvements of the student models.

D. Compare With Other Distillation Frameworks

We compared the proposed NKDFD with 13 methods: AT [35], FT [29], OFD [30], AB [36], CWD [20], FBKD [38], PKD [21], MGD [22], ReviewKD [31], CKA [32], DIST [33], FGD [37], and MasKD [34]. Uniform feature distillation methods (FT, OFD, ReviewKD, CKA, DIST, and PKD) do not consider spatial heterogeneity and uniformly process foreground/background features. Nonuniform approaches (AT,

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TABLE IV
GENERALIZATION PERFORMANCE EVALUATION OF DETECTION MODELS WITH NKDFD DISTILLATION ON VHR10 DATASET

Class	Sub-Class	Method	mAP (%)	mAP_{50} (%)	mAP_{75} (%)	mAP_s (%)	mAP_m (%)	mAP_l (%)	$Params$ (M)	$FLOPs$ (G)
Single Stage Detector	Anchor-based	YOLOF(T)	32.4	65.7	28.9	0.0	25.4	45.5	61.539	45.791
		YOLOF(S)	15.8	40.3	10.6	0.0	10.8	23.5	16.470	10.856
		YOLOF(ours)	**20.2**	**48.0**	**15.0**	0.0	**12.5**	**30.6**	**16.470**	**10.856**
		GFL(T)	51.4	88.6	55.7	29.3	44.8	56.6	51.283	64.521
		GFL(S)	46.8	83.9	50.7	24.5	42.1	52.2	19.271	35.712
		GFL(ours)	**49.2**	**86.0**	**53.1**	**40.0**	**44.4**	**57.6**	**19.271**	**35.712**
		PAA(T)	48.8	88.4	49.4	37.6	42.6	57.7	51.112	63.686
		PAA(S)	40.1	78.7	37.3	24.4	32.6	51.6	19.100	34.877
		PAA(ours)	**41.3**	**81.8**	35.9	16.7	**35.3**	50.7	**19.100**	**34.877**
		ATSS(T)	50.3	88.9	52.6	25.5	45.5	58.2	51.112	63.686
		ATSS(S)	44.0	82.7	41.1	15.6	36.9	54.1	19.100	34.877
		ATSS(ours)	**44.7**	**84.4**	**42.6**	**26.7**	**39.8**	**55.2**	**19.100**	**34.877**
	Anchor-free	VFNet(T)	50.8	89.7	50.0	36.7	45.3	58.1	51.709	60.825
		VFNet(S)	38.8	82.9	29.8	33.5	36.2	44.0	19.697	32.017
		VFNet(ours)	**43.8**	**83.1**	**38.7**	15.6	**39.9**	**50.2**	**19.697**	**32.017**
		TOOD(T)	51.7	89.6	55.0	37.1	44.9	60.4	51.017	62.745
		TOOD(S)	46.6	86.4	45.3	26.7	41.4	53.6	19.006	33.937
		TOOD(ours)	**50.0**	**87.9**	**51.4**	**28.3**	**44.9**	**55.5**	**19.006**	**33.937**
		FCOS(T)	31.0	72.2	19.5	13.3	25.6	43.6	51.114	63.686
		FCOS(S)	25.5	62.1	14.6	0.0	20.9	31.6	19.103	34.877
		FCOS(ours)	**28.1**	**64.3**	**21.4**	**6.7**	**23.6**	**32.8**	**19.103**	**34.877**
		YOLOX(T)	49.5	83.7	53.4	36.3	44.9	52.7	54.155	33.898
		YOLOX(S)	36.5	72.8	33.7	12.5	35.9	33.1	5.035	3.202
		YOLOX(ours)	**39.1**	**75.0**	**37.8**	10.0	35.2	**43.3**	**5.035**	**3.202**
		CenterNet(T)	41.2	84.5	33.3	23.3	38.2	51.2	51.110	63.675
		CenterNet(S)	38.3	79.9	33.2	26.7	34.2	45.7	19.098	34.866
		CenterNet(ours)	**40.3**	**82.6**	**34.1**	10.0	**38.1**	**48.6**	**19.098**	**34.866**
Two Stage Detector		FRCNN(T)	49.7	89.4	51.2	31.7	48.8	51.4	60.356	82.642
		FRCNN(S)	43.3	82.1	40.8	24.2	39.5	43.1	28.326	49.746
		FRCNN(ours)	**46.0**	**84.4**	**46.4**	**34.2**	**41.8**	**47.8**	**28.326**	**49.746**

AB, FGD, FBKD, and CWD) ensure a focus on specific regions by using attention mechanisms [see Fig. 2(a)]. MasKD achieves nonuniform distillation using spatial feature decoupling masks, whereas MGD enhances feature generation via teacher masking, balancing foreground and background losses [see Fig. 2(b)]. All methods used RetinaNet/ATSS frameworks with ResNet-101/ResNet-18 as teacher/student backbones and identical configurations for a fair comparison of the DOTA and RSOD datasets.

Table V presents the results on the DOTA dataset. For RetinaNet, DIST provided the best performance among uniform distillation methods (+6.6% mAP and +8.5% mAP_{50}). NKDFD surpassed all nonuniform methods with superior gains (+7.8% mAP, +10.9% mAP_{75}, +6.8% mAP_m, and +10.1% mAP_l). Other nonuniform methods exhibited limited gains or degradation (e.g., MGD: +4.9%; CWD: −5.4%; and AT: −0.8%) due to inadequate feature decoupling. NKDFD had an mAP_{50} comparable to DIST Δ 0.1% and ranked second in mAP_s (after ReviewKD).

For ATSS, all methods improved the accuracy. NKDFD achieved the maximum mAP (48.8%) with +4.4% mAP_{50} (second to PKD) and +0.9% mAP_{75} over DIST. It provided the optimum medium-scale detection performance (+6.3% mAP_m) and was competitive for small and large-scale detection (13.7% mAPs, 63.2% mAP_l).

The experimental results on the RSOD dataset (see Table VI) demonstrate that the NKDFD distillation framework consistently achieved the optimum performance based on various evaluation metrics. Notably, it attained state-of-the-art mAP scores of 44.8% with RetinaNet and 54.0% with ATSS.

To evaluate the distillation performance under different spatial coupling conditions of the features, we conducted experiments based on the NKDFD framework. The results are presented in Table VII. In terms of overall detection accuracy, the Gaussian-based mode achieves the highest mAP of 37.7%, surpassing both the coupled-based (35.7%) and binary-based (37.4%) methods. It also excels under stricter localization criteria, attaining the best mAP_{50} (60.6%) and mAP_{75} (40.4%) results. Regarding scale sensitivity, the Gaussian-based approach shows superior performance on medium and large targets compared with the coupled-based method, though it is slightly less effective than the binary-based mode on small

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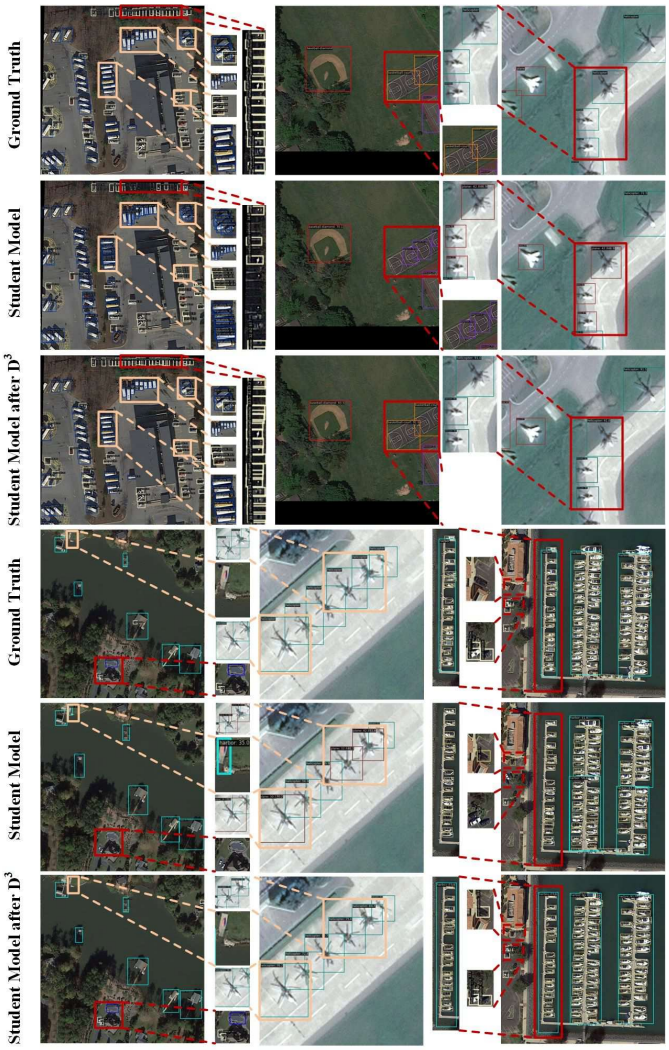


Fig. 7. Detection performance of Tood-R18 model on the DOTA dataset before and after applying the NKDFD framework.

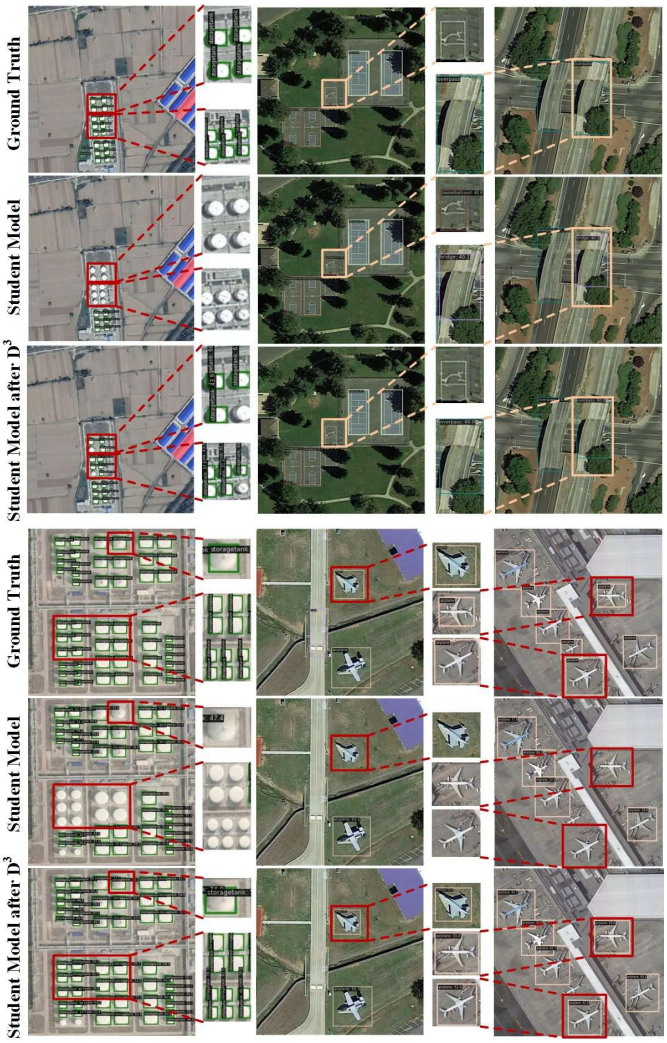


Fig. 8. Detection performance of Tood-R18 model on the DIOR dataset before and after applying the NKDFD framework.

targets, likely due to the retention of redundant information under soft decoupling. Overall, the Gaussian-based mode demonstrates competitive or superior performance across most metrics, particularly in overall accuracy and localization quality, validating its ability to balance feature separation and information retention for enhanced robustness in complex scenarios.

To further compare the effects of different nonlinear representation modules—namely, CNN/MLP, Transformer, and KAN—on the distillation performance of the NKDFD module, we conducted a comparative study using the RetinaNet framework. The experimental results are summarized in Table VIII.

The experimental results indicate that the architecture of the semantic decoupling module has a significant impact on distillation performance. Specifically, the KAN-based module achieved the best overall performance, with an mAP of 37.7%, outperforming its CNN- and Transformer-based counterparts by 1.1 and 0.6 percentage points, respectively. It is noteworthy that the KAN module attained a mAP_{75} of 40.4%, substantially surpassing the other architectures. This demonstrates that its stronger nonlinear representation capability leads to superior distillation performance.

E. Ablation Study

We conducted an ablation study on the DOTA dataset using RetinaNet as the remote sensing object detection framework to evaluate the effectiveness of the proposed modules. We evaluated the impact of the GDSD, KSDS, and CFI modules on KD using a stepwise combination approach. As shown in Table IX, incorporating GDSD improved all metrics, with a 0.7% mAP increase. The sequential incorporation of the CFI and KSDS modules provided mAP improvements of 1.1% and 0.9%, respectively. The student model with the three modules achieved a 1.8% mAP improvement over the baseline, demonstrating the effectiveness of the modules and their synergistic effects. This result confirms that the proposed framework enhances KD performance in remote sensing object detection using a modular design and cooperative optimization.

Meanwhile, we conducted ablation studies on the DOTA for the anchor-free model—CenterNet—and the two-stage

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TABLE V

PERFORMANCE COMPARISON OF VARIOUS KD METHODS ON THE DOTA DATASET USING RETINANET AND ATSS FRAMEWORKS

Method			RetinaNet						ATSS					
			mAP	mAP_{50}	mAP_{75}	mAP_s	mAP_m	mAP_l	mAP	mAP_{50}	mAP_{75}	mAP_s	mAP_m	mAP_l
Teacher Model			43.8	66.3	46.1	6.3	42.4	67.3	54.4	76.6	61.3	18.5	54.8	70.8
Student Model			29.9	52.2	29.5	3.0	28.7	46.5	43.6	67.7	47.7	9.6	43.9	58.1
Uniform Knowledge Distillation (Fig. 2a)		FT	20.2	40.5	18.0	1.2	17.6	35.8	48.6	72.0	54.3	13.4	49.7	63.6
		OFD	28.2	49.8	28.0	3.3	28.6	41.1	45.4	68.5	50.1	12.7	47.2	57.6
		ReviewKD	35.9	58.3	37.9	**4.1 (+1.1)**	34.9	55.0	47.3	70.8	52.5	11.8	48.8	61.6
		CKA	31.3	53.9	31.2	3.1	29.9	48.9	47.1	70.7	52.4	13.3	48.4	61.1
		DIST	36.5	**60.7 (+8.5)**	38.0	4.0	33.7	56.5	48.6	72.3	54.0	**14.0 (+4.4)**	48.9	**63.8 (+5.7)**
		PKD	35.5	59.2	37.0	3.9	32.3	55.9	48.4	**72.4 (+4.7)**	54.0	13.9	48.7	63.1
Non-uniform Knowledge Distillation	Fig. 2b	AT	29.1	51.2	28.3	2.8	27.5	46.0	48.5	71.6	53.8	13.9	49.5	63.6
		AB	30.1	52.8	29.6	3.0	28.6	45.4	46.8	70.4	52.4	13.4	47.9	60.9
		FGD	33.1	55.5	34.3	4.3	34.6	47.0	44.7	67.0	49.6	13.6	47.8	55.5
		FBKD	30.2	52.9	29.9	3.2	29.0	47.2	47.3	70.6	52.8	11.9	48.6	61.8
		CWD	24.5	44.4	23.4	2.2	22.9	37.6	44.2	66.4	49.2	13.2	47.9	54.0
	Fig. 2b	MasKD	34.4	56.7	35.8	4.1	34.3	51.5	47.0	68.8	52.6	14.1	50.8	58.7
		MGD	34.8	58.2	36.0	3.7	32.9	54.3	47.2	70.9	52.3	13.1	48.5	61.5
	Fig. 2c	NKDFD	**37.7 (+7.8)**	60.6	**40.4 (+10.9)**	3.9	35.5 (+6.8)	**56.6 (+10.1)**	**48.8 (+5.2)**	72.1	**54.9 (+7.2)**	13.7	**50.2 (+6.3)**	63.2

TABLE VI

PERFORMANCE COMPARISON OF VARIOUS KD METHODS ON THE RSOD DATASET USING RETINANET AND ATSS FRAMEWORKS

Method			RetinaNet						ATSS					
			mAP	mAP_{50}	mAP_{75}	mAP_s	mAP_m	mAP_l	mAP	mAP_{50}	mAP_{75}	mAP_s	mAP_m	mAP_l
Teacher Model			41.8	80.1	41.6	0.6	38.1	52.7	57.5	91.9	61.6	18.5	61.0	64.6
Student Model			35.8	76.3	34.0	0.3	29.8	47.2	52.6	87.4	59.7	13.3	58.2	60.0
Uniform Knowledge Distillation (Fig. 2a)		FT	36.1	76.5	34.3	0.2	29.6	47.6	50.7	85.3	57.1	13.4	56.8	58.5
		OFD	41.5	76.8	41.6	0.5	33.7	52.7	50.1	81.2	54.6	15.3	59.9	57.1
		ReviewKD	39.8	78.9	37.9	0.7	32.7	50.0	49.2	87.0	56.6	13.5	57.9	57.0
		CKA	39.6	79.8	37.7	0.4	29.9	50.3	53.3	89.7	56.5	13.1	58.5	60.9
		DIST	42.9	79.5	43.4	0.5	35.2	53.9	52.6	90.4	58.2	**16.6 (+3.2)**	**60.2 (+2.0)**	58.8
		PKD	41.6	78.3	42.6	0.3	32.6	52.4	53.6	**92.0 (+4.6)**	57.8	13.1	56.6	**61.1 (+1.1)**
Non-uniform Knowledge Distillation	Fig. 2b	AT	36.3	76.4	34.5	0.3	30.3	47.3	52.7	86.8	57.5	13.7	57.2	60.6
		AB	38.2	80.3	33.7	**2.4 (+2.1)**	**47.2 (+17.4)**	46.0	51.0	90.6	52.2	13.7	57.2	60.6
		FGD	39.0	78.4	34.6	0.6	33.7	49.3	52.2	80.7	57.9	14.2	59.8	59.6
		FBKD	40.3	79.4	38.6	0.7	32.4	50.5	52.9	89.3	55.7	8.8	52.5	60.8
		CWD	42.5	79.5	42.4	0.7	39.0	52.4	52.3	84.3	59.4	15.3	59.6	59.2
	Fig. 2b	MasKD	35.1	76.8	31.3	0.2	32.6	45.6	51.8	81.0	59.0	14.7	60.1	59.1
		MGD	43.5	78.1	42.7	0.5	34.3	53.6	52.9	90.1	57.8	11.7	58.0	60.8
	Fig. 2c	NKDFD	**44.8 (+9.0)**	**85.4 (+9.1)**	**47.3 (+13.3)**	1.3	39.7	**54.4 (+7.2)**	**54.0 (+1.4)**	89.9	**59.7 (+0.0)**	11.8	53.0	**61.1 (+1.1)**

TABLE VII

DISTILLATION PERFORMANCE OF THE RETINANET UNDER DIFFERENT SPATIAL FEATURE COUPLING MODES

Method	mAP	mAP_{50}	mAP_{75}	mAP_s	mAP_m	mAP_l
Coupled	35.7	58.4	37.6	4.1	35.0	54.9
Binary	37.4	60.1	39.6	4.7	36.2	58.1
Gaussian	37.7	60.6	40.4	3.9	35.5	56.6

TABLE VIII

DISTILLATION PERFORMANCE OF THE NKDFD FRAMEWORK WITH DIFFERENT MODULES ON RETINANET

Method	mAP	mAP_{50}	mAP_{75}	mAP_s	mAP_m	mAP_l
CNN/MLP	36.6	59.3	38.4	4.1	35.2	56.3
Transformer	37.1	60.1	38.5	4.5	36.1	56.8
KAN	37.7	60.6	40.4	3.9	35.5	56.6

model—FRCNN, with the corresponding results presented in Tables X and XI. These results confirm that the modules introduced in NKDFD are effective and complementary across different detection paradigms. They enhance the student model's capability to detect objects in remote sensing images, demonstrating strong generalizability and architectural compatibility.

To intuitively compare the impact of the σ factor on the performance of the NKDFD framework, we fixed the value of σ and conducted experiments under different

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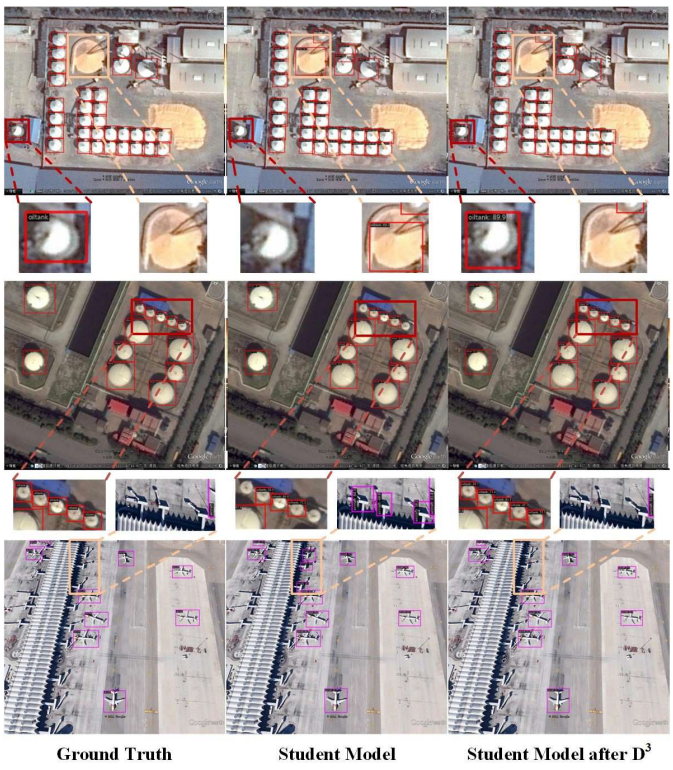


Fig. 9. Detection performance of Tood-R18 model on the RSOD dataset before and after applying the NKDFD framework.

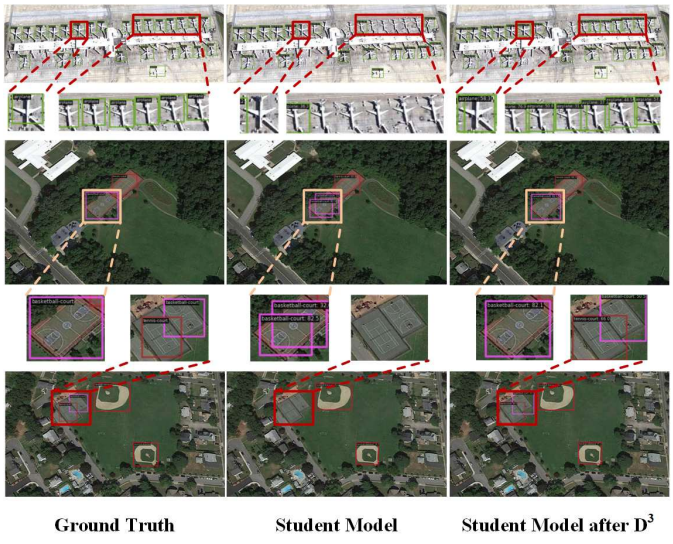


Fig. 10. Detection performance of Tood-R18 model on the VHR10 dataset before and after applying the NKDFD framework.

settings to demonstrate its influence. Specifically, we varied the balancing parameter σ from 0 to 1 in increments of 0.1, generating multiple configurations to assign different weight ratios to the foreground and background features. The corresponding ablation results are presented in Table XII.

The experimental results demonstrate that the value of σ has a significant impact on distillation performance. Among the fixed settings, both relatively small and large values of σ can lead to competitive detection accuracy. For instance, the mAP obtained with $\alpha = 0.2$ is only slightly lower than that achieved with $\alpha = 0.9$, which aligns with the findings in [27], indicating that using background features

TABLE IX

ABLATION STUDY ON DIFFERENT MODULES OF THE NKDFD FRAMEWORK USING RETINANET ON THE DOTA DATASET

Module			Accuracy Evaluation Metrics					
GDSD	KSDS	CFI	mAP	mAP_{50}	mAP_{75}	mAP_s	mAP_m	mAP_l
			35.9	58.3	37.9	4.1	34.9	55.0
✓			36.6	59.1	38.7	4.2	35.3	56.9
✓	✓		36.8	59.2	39.3	4.5	35.5	57.5
✓		✓	37.0	59.6	38.8	4.2	35.8	57.2
✓	✓	✓	37.7	60.6	40.4	3.9	35.5	56.6

TABLE X

ABLATION STUDY ON DIFFERENT MODULES OF THE NKDFD FRAMEWORK USING CENTERNET ON THE DOTA DATASET

Module			Accuracy Evaluation Metrics					
GDSD	KSDS	CFI	mAP	mAP_{50}	mAP_{75}	mAP_s	mAP_m	mAP_l
			38.4	63.7	41.1	10.0	38.4	49.9
✓			40.9	66.2	45.0	10.4	40.7	54.4
✓	✓		41.7	65.4	46.4	11.1	41.7	54.5
✓		✓	40.1	63.4	44.4	10.8	40.6	51.4
✓	✓	✓	44.5	68.0	50.1	13.2	45.3	56.4

TABLE XI

ABLATION STUDY ON DIFFERENT MODULES OF THE NKDFD FRAMEWORK USING FRCNN ON THE DOTA DATASET

Module			Accuracy Evaluation Metrics					
GDSD	KSDS	CFI	mAP	mAP_{50}	mAP_{75}	mAP_s	mAP_m	mAP_l
			40.8	65.2	45.1	6.3	43.9	54.9
✓			42.0	65.7	46.9	6.1	45.9	56.7
✓	✓		42.0	66.6	46.6	7.1	45.4	57.0
✓		✓	42.2	66.2	46.4	6.7	45.8	57.2
✓	✓	✓	42.4	66.0	47.5	6.4	46.1	56.7

TABLE XII

DISTILLATION PERFORMANCE COMPARISON OF THE NKDFD FRAMEWORK WITH RETINANET ACROSS DIFFERENT σ FACTORS (ADP MEANS ADAPTIVE)

Method	mAP	mAP_{50}	mAP_{75}	mAP_s	mAP_m	mAP_l
0.1	37.3	60.2	39.6	4.7	36.2	56.7
0.2	37.6	60.1	39.8	4.6	36.9	57.2
0.3	36.2	59.1	38.4	4.5	35.8	54.4
0.4	36.6	59.1	38.1	4.3	35.9	55.9
0.5	36.7	59.4	38.7	4.3	36.1	56.1
0.6	37.3	59.9	39.7	4.4	36.4	56.7
0.7	36.2	58.3	38.2	4.5	35.8	54.0
0.8	37.3	59.9	39.4	4.3	36.0	57.3
0.9	37.5	60.1	39.7	4.7	36.5	57.8
Adp	37.7	60.6	40.4	3.9	35.5	56.6

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TABLE XIII
QUANTITATIVE METRICS FOR EVALUATING FEATURE CONSISTENCY AND DISPARITY ACROSS DIFFERENT COMBINATIONS

	Group	Linear CKA	RBF Kernel CKA	PCKA	CCA	Robust CCA	SVD + Procrustes	PCA (2)+ Procrustes	PCA (10)+ Procrustes
Random Data1 vs Random Data1	1	1.000	1.000	1.000	1.000	0.505	0.000	0.000	0.000
Random Data1 vs Random Data2	2	0.496	0.654	0.485	0.040	0.034	35.064	34.304	34.796
MF(FF) vs Random Data	3	0.350	0.463	0.314	0.080	0.038	35.080	34.455	34.568
MB(FB) vs Random Data	4	0.443	0.562	0.410	0.076	0.037	35.063	35.114	34.868
MB(FF) vs Random Data	5	0.352	0.464	0.314	0.078	0.037	35.071	34.528	34.788
MF(FF) vs MB(FF)	7	0.517	0.772	0.422	0.188	0.095	34.971	10.811	25.812
MB(FB) vs MF(FB)	8	0.493	0.614	0.511	0.125	0.063	34.913	14.175	26.534
MF(FF) vs MF(FB)	9	0.402	0.508	0.384	0.291	0.147	32.765	32.340	33.842
MB(FB) vs MB(FF)	10	0.403	0.509	0.390	0.295	0.149	32.738	32.315	34.345
MF(FF) vs MB(FB)	11	0.358	0.482	0.334	0.104	0.053	35.049	32.361	35.544
MF(FB) vs MB(FF)	12	0.354	0.477	0.331	0.104	0.053	35.054	32.671	34.363

alone can yield detection performance comparable to using only foreground features. The adaptive weighting method ultimately achieved the best overall results, outperforming all fixed-weight configurations in terms of mAP (37.7%), mAP_{50} (60.6%), and mAP_{75} (40.4%), demonstrating its ability to dynamically adjust weights according to image content. Overall, the results validate the importance of the proposed adaptive balancing mechanism in coordinating the transfer of foreground and background knowledge.

F. Discussion: Feature Consistency Analysis

We utilized a spatial mask based on a 2-D Gaussian distribution for feature space decoupling, implementing independent adapters for foreground and background features. The experiments indicate that considering spatial heterogeneity in remote sensing object detection and analyzing the output characteristics of different attribute features processed by independent adapters are crucial. Attribute-based spatial decoupling was used to obtain foreground features (FF and background features (FB). We derived the foreground feature adapter (H_s^{frg}) and the background feature adapter (H_s^{bkg}), using the student model (FCOS) on the DOTA dataset as an example. The foreground and background features were input into different adapters, generating four output features: H_s^{frg}(FF), H_s^{frg}(FB), H_s^{bkg}(FB), and H_s^{bkg}(FF). Feature consistency was evaluated through pairwise combinations of these outputs, demonstrating the necessity of independent adaptation after spatial decoupling.

We quantified feature relationships through consistency and discrepancy metrics. The CKA [66] was used to assess consistency. It captures linear/nonlinear correlations (using linear CKA, RBF CKA, and PCKA). CCA was used to identify the maximum correlation direction (standard CCA for linearity, robust CCA for noise resistance). Procrustes analysis was used to assess discrepancy. It performs shape analysis using geometric transformations. Singular value decomposition (SVD) and principal component analysis (PCA) (with 2/10 principal components) were used for dimensionality reduction to balance computational efficiency and feature preservation. Higher consistency and lower discrepancy indicate higher feature similarity.

Table XIII shows the consistency/discrepancy metrics for two primary groups: control groups based on random data and experimental groups with different output feature combinations. The results indicate that adapter alignment is essential. The FF processed by H_s^{frg} versus H_s^{bkg} show marginally higher consistency than random features but significantly lower consistency than identical features (Group 7). Mismatched processing yields ineffective transformation (Group 9 $\approx$ Groups 3/6), and cross-adapter combinations produce random-equivalent metrics, confirming high correlations between adapters and features.

The experimental results confirm that the feature-adapter design must align with the feature attributes. When the FF were processed by H_s^{frg} versus H_s^{bkg}, their consistency metrics slightly exceeded those of random feature combinations but were significantly lower than those of identical feature combinations (e.g., Group 7). The discrepancy metrics approximate those of random data, validating the specificity of adapter attributes. Similarly, processing the FB using mismatched adapters resulted in consistency/discrepancy metrics equivalent to random combinations, requiring an attribute-specific adapter design. For identical adapters processing mismatched attributes (e.g., Groups 9 and 10), the consistency metrics had lower values than those of random combinations with comparable discrepancy metrics, indicating an ineffective transformation, and the outputs were indistinguishable from random data (e.g., Group 9 $\approx$ Groups 3/6 and Group 10 $\approx$ Groups 4/5). Cross-adapter combinations (nonmatching features processed by opposing adapters) exhibited metrics identical to random data, confirming strong feature-adapter correlations.

The visualizations of the feature consistency metrics listed in Table XIII are presented in Fig. 11. The consistency and disparity metrics for random data combinations (Group 2) are shown in red. Except for the identical data pairs (Group 1), the consistency metrics for the other combinations are slightly higher or comparable to those of random data pairs (Group 2). This result indicates a strong correlation between feature and

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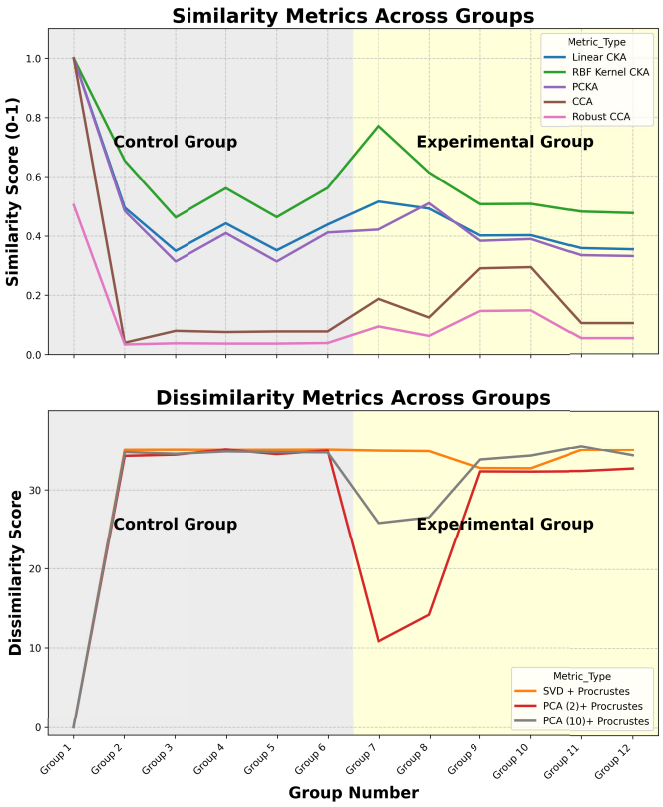


Fig. 11. Comparison of feature consistency and discrepancy evaluation metrics across different combinations.

adapter attributes, particularly in remote sensing images with pronounced spatial heterogeneity, where a decoupled adapter design is necessary to obtain distinct attribute features to ensure effective distillation.

V. Conclusion

This article proposed the NKDFD framework to address spatial heterogeneity, semantic discrepancy, and scale diversity in remote sensing object detection using KD. Its core innovation is the decoupling of features in the foreground and background. The method performs independent nonuniform feature extraction and knowledge transfer to mitigate potential discrepancies in detection patterns between teacher and student models caused by high spatial heterogeneity (e.g., over-reliance on background features, foreground features, or suboptimal weighted combinations). The NKDFD framework includes a KAN-based KSDS module and a CFI module for nonuniform feature adaptation to address semantic discrepancy and scale diversity. Experiments on the DOTA, DIOR, RSOD, and VHR10 remote sensing datasets demonstrated that NKDFD achieved high detection accuracy with a compact model size. It outperformed existing uniform and nonuniform KD methods and improved the performance of student models. We observed a high correlation between feature attributes and adapters. Nonuniform feature extraction is essential. Mismatched adapter-attribute combinations produced outputs statistically indistinguishable from randomly generated data. Thus, independent adapters for foreground and background features are required to optimize the KD performance.

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